

Jonathan Jacob Koshy

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Education

University of Waterloo
BASc. Electrical Engineering

Waterloo, ON
Expected 2029

Experience

Embedded Software Intern – Belimo - Montreal, QC May 2026 – Aug 2026

- Developed bare metal **Embedded C** on **8 bit PIC18 MCU**, reducing HVAC controller memory use by **20%**
- Validated **CAN, UART, and I2C** signals with lab tools to isolate timing faults and controller state bugs
- Optimized interrupt timing and debounce windows, improving keypad navigation latency by **30%**
- Implemented **ADC digital filtering** for gas sensing firmware, improving measurement accuracy by **10%**
- Improved **BACnet** communication by debugging UART parsing, frame handling, and state recovery faults
- Tested firmware against UL verification guidelines across sensor behavior, edge cases, and PCB level validation

ASIC Digital - UWASIC – Waterloo, ON Jan 2026 – Present

- Architected a 64-bit **Ethernet parser** with pipelined header extraction, CRC validation, and frame buffering
- Designed an **ARP request validator** to verify Ethernet/IPv4 fields and match target IPs for hardware replies
- Implemented an SPI controlled PWM peripheral with Mode 0 SPI, CDC logic, and Cocotb tests

Embedded Software Intern – AeroCardia - Montreal, QC Sep 2025 – Dec 2025

- Developed **C++ sensor** drivers for IMU, PPG, and temperature sensors, enabling **50 Hz** biosignal capture
- Built **RTOS** pipeline using queues to buffer biosensor data before **BLE** transmission, resulting in **<1%** packet loss
- Optimized sensor sampling and buffering paths, reducing dropped samples by **25%** during continuous monitoring
- Implemented secure **BLE OTA** updates with image verification and rollback, cutting update time to **under 1 min**
- Designed an O2 sensor PCB in **Altium** with onboard EEPROM for persistent sensor calibration data

Embedded Software Member – UWARG – Waterloo, ON Dec 2024 – Sep 2025

- Implemented roll and yaw motor mixing firmware, mapping pilot inputs to motor outputs in fixed-wing modes
- Tuned **PID** attitude loops, reducing manual flight oscillation by **20%** during stabilization tests
- Improved control loop timing and actuator output scaling, reducing motor command jitter by **30%**

Embedded Software Intern – Electrium Mobility – Waterloo, ON May 2025 – Aug 2025

- Built **ESP32** firmware for an electric skateboard dashboard, streaming **BLE** telemetry using FreeRTOS
- Integrated **VESC** motor controller data over **UART**, streaming RPM, current, and battery metrics
- Added packet parsing and checksum validation for skateboard diagnostics, reducing malformed frames by **25%**

Projects

Edge AI Dementia Companion (HackCanada Winner) *Python, YOLOv8, OpenCV, TypeScript, RPi 5* 

- Built an edge AI memory aid with RPi 5 & Hailo-8L using YOLO to track missing objects and last seen locations
- Added medication checks, object memory, and WebSocket snapshots for phone based caregiver assistance

ESP32 Wi-Fi Media Controller | *C++, ESP32, LVGL, Wi-Fi, REST APIs* |  | 

- Built a custom handheld media dashboard with **LVGL** for PC metrics, media state, and Discord data
- Designed a TCP JSON protocol with queued updates, RGB565 artwork rendering, and touchscreen media controls

RISC-V Interrupt Driven Embedded Controller | *RISC-V Assembly, SiFive FE310, Interrupts*

- Built an interrupt driven **RISC-V** controller using GPIO ISRs to capture pseudo random input events
- Configured **PLIC, GPIO, mtvec, mie**, and **mstatus** registers for low level interrupt handling

Skills

- **Programming:** C, C++, Python, RISC-V, MATLAB, Git
- **Protocols/Platforms:** SPI, I2C, UART, BLE, CAN, Ethernet, PWM, RTOS, STM32, ESP32, Raspberry Pi, Linux
- **Lab Tools:** Oscilloscopes, Logic Analyzers, Multimeters, Soldering Irons, Bench Power Supplies
- **HDL/Digital Design:** Verilog, RTL Design, FSMs, CDC, ASIC, FPGA